LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 11 Test: Electricity

Chapter / Topic: Electricity

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. In numericals, write the formula, substitute values with units, and give the final answer with its unit.

SECTION A

6 marks

Choose the correct option.

Q1. A wire of resistance R is cut into five equal parts, which are then connected in parallel. If the equivalent resistance is R′, the ratio R/R′ is: **[1]**

(a) 1/25 (b) 1/5
(c) 5 (d) 25

Q2. Which of the following does NOT represent electric power in a circuit? **[1]**

(a) I^2R (b) IR^2
(c) VI (d) V^2/R

Q3. The SI unit of electric charge is the: **[1]**

(a) ampere (b) volt
(c) coulomb (d) ohm

Q4. How much current does an 1100 W appliance draw from a 220 V supply? **[1]**

(a) 0.5 A (b) 2 A
(c) 5 A (d) 10 A

Q5. An electric bulb is rated 220 V, 100 W. When it is operated on 110 V, the power consumed is: **[1]**

(a) 25 W (b) 50 W
(c) 75 W (d) 100 W

Q6. **Assertion (A):** An ammeter is always connected in series with the component whose current is to be measured.
Reason (R): The same current flows through all components connected in series. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. How much work is done in moving a charge of 2 C across two points that have a potential difference of 12 V? **[2]**

OR

Why are the coils of electric toasters and electric irons made of an alloy rather than a pure metal?

Q8. Resistors of 4 Ω and 6 Ω are connected in series to a 12 V battery. Find the current in the circuit and the potential difference across the 4 Ω resistor. **[2]**

SECTION C

6 marks

Q9. A copper wire has a diameter of 0.5 mm and a resistivity of 1.6×10^{-8} Ω m. **[3]**

(i) What length of this wire will have a resistance of 10 Ω? **[2]**

(ii) If the diameter is doubled (same length), how does the resistance change? **[1]**

Q10. Three resistors of 5 Ω, 10 Ω and 30 Ω are connected in parallel to a 12 V battery. **[3]**

(i) Find the equivalent resistance. **[1]**

(ii) Find the current through each resistor. **[1]**

(iii) Find the total current drawn from the battery. **[1]**

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) State Joule's law of heating. **[1]**

(ii) An electric iron of resistance 20 Ω takes a current of 5 A. Calculate the heat produced in 30 s. **[2]**

(iii) Two lamps, one rated 100 W at 220 V and the other 60 W at 220 V, are connected in parallel to a 220 V supply. What current is drawn from the supply? **[2]**

OR

(a) Why is a parallel arrangement used in domestic wiring rather than a series arrangement? Give two reasons. [2]

(b) An electric refrigerator rated 400 W runs 8 hours a day. What is the cost of the energy it uses in 30 days at ₹3.00 per kWh? [3]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. In an experiment, a student measured the current (I) through a resistor for different potential differences (V). The readings were: I = 0.5, 1.0, 2.0, 3.0, 4.0 A and V = 1.6, 3.4, 6.7, 10.2, 13.2 V respectively. **[4]**

(i) What relationship between V and I do these readings show? **[1]**

(ii) Calculate the resistance using the reading I = 2.0 A. **[1]**

(iii) The resistor is replaced by a wire of the same material and thickness but double the length. What current will flow at 6.7 V? **[2]**

~ All the best! ~